Experimental Determination of Poisson's Ratio for Asphalt Binder

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Introduction

The materials used in the construction of asphalt pavements undergo large amount of stress, from temperature changes, to millions of loads applied by trucks and other vehicles that travel on top of the road. These loads create damage that accumulates over time and eventually lead to cracking and permanent deformation. As cracks grow, potholes form and the serviceability of the roads is severely diminished.

Understanding the mechanical properties of roadway materials allows for better management of crack formation and other failures associated with asphalt (rutting, pothole formation, etc.). A key mechanical property is Poisson's ratio. Poisson's ratio is related to the bulk modulus, shear modulus, and Young's modulus, and it is used to determine how the material behaves under various mechanical loading configurations.

Experimental determination of the Poisson's ratio is difficult and for asphalt binders this value is assumed. This causes uncertainties when predicting material behavior, a critical component of asphalt pavements design. By experimentally determining this ratio, more rigorous models can be developed to predict the failure of asphalt pavements.

Poisson's Ratio

Poisson's ratio describes the resistance of a material to distort under mechanical load rather than to alter in volume (Greaves et. al, 2011). The ratio represents the relationship between the strain in the horizontal direction (ε_x) and the corresponding strain in the vertical direction(ε_y).

An example is shown in Figure 1. When a compressive force is applied in the vertical y-direction, the material contracts in the vertical direction, which is identified by using a negative sign, whereas, in the horizontal direction, the material expands, which is positive. This is why the negative sign is used in Equation 3.

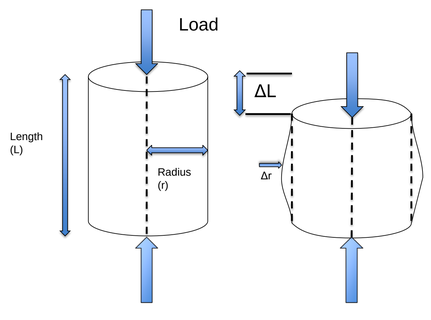


Figure 1 – Interpretation of Poisson's Ratio

(1) $$e_x = \frac{\Delta r}{r_o}$$

(2) $$e_y = \frac{\Delta L}{L_o}$$

(3) $$v = -\frac{e_x}{e_y}$$

Another example is shown in Figure 2; in this case, a compressive force is applied to a cylindrical specimen along the vertical diameter and the cylinder contracts along the vertical diameter and expands along the horizontal diameter.

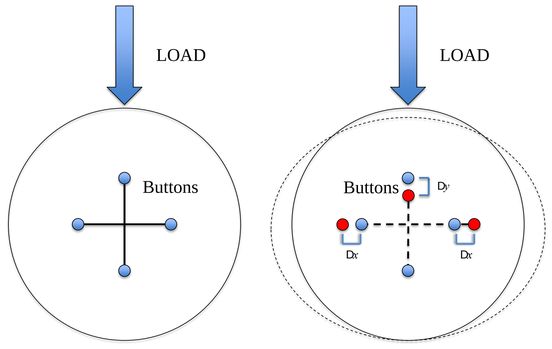


Figure 2 – IDT Testing Method for determining Poisson's Ratio

This setup is used in the current AASHTO specification to obtain, creep compliance, strength and Poisson ratio of asphalt mixtures.

Methodology

By making slight alterations to AASHTO specification, it is possible to perform IDT creep tests on asphalt binder cylindrical specimens and to determine Poisson's ratio. To prepare the binder specimens, a cylindrical container (75mm in height, 150 in diameter) was designed and used to mold the asphalt binder samples. The heated binder was poured into the container, allowed to cool down at room temperature, and then placed in a freezer to prevent any viscous flow and preserve the shape of the specimen. Using this procedure, three cylindrical specimens were prepared.





Once the specimen hardened, it was removed from the container, and placed back in the freezer until it was ready to be tested.

To perform the tests, a servo-hydraulic MTS loading frame was used together with an environmental chamber. Each specimen was tested at two temperatures: -18°C and -24°C. A load was instantaneously applied and then kept constant over the entire duration of the test. Horizontal and vertical deformations were recorded on both sides of each sample, along with the axial force that was applied throughout the test. In order to record deformations during the test, extensometers were attached to both sides of each specimen.

Four metallic buttons were glued to both sides of the samples to position the extensometers to measure the horizontal and vertical deformations in the middle portion of the specimens. Refer to Fig. 2 for the orientation of the buttons, and the extensometers, which are represented by the perpendicular lines. The dotted lines represent the deformation after the load is applied.



Results for PAV Binder 58-34 @ -18°C

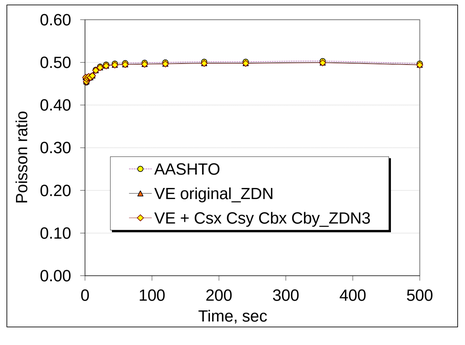


Figure 3 – Sample 3 tested at -18°C

Three different methods were used in the determination of Poisson's ratio. As seen in Figure 3, all three methods show a value of approximately 0.5 for Poisson's ratio when the binder is tested at -18°C.

Results for PAV Binder 58-34 @ -24°C

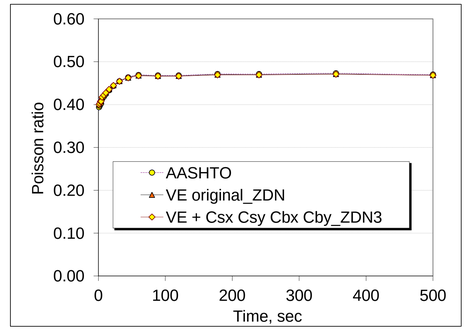


Figure 4 – Sample 2 tested at -24°C

When the test was performed at lower temperatures, the value of Poisson's ratio was slightly lower. The three methods produced similar values, approximately 0.47 throughout the testing while at -24°C.

Conclusion

Three samples of PAV binder of performance grade 58-34 were subjected to an indirect tensile creep test in order to determine Poisson's ratio for the binder. Several conclusions can be drawn:

- At -18°C, the value of Poisson's ratio is approximately 0.5, which is the value assumed in calculations over a wide range of temperatures.
- At -24°C, however, the value of Poisson's ratio decreases to 0.47, which indicates a slight dependence on temperature.
- IDT configurations appears to be a viable test method to determine Poisson's ratio for asphalt binders.

Experimental determination of Poisson's ratio has resulted in the validation of the estimated value that has been used in the design of roads. With this knowledge, models of road failure can be developed in a more accurate sense. This allows for more reliability in predicting the performance of asphalt pavements.

References

1. American Association of State Highway and Transportation Officials T322-03-UL. (2006). *Standard Method of Test for Determining the Creep Compliance and the Strength of Hot-Mix Asphalt (HMA) Using the Indirect Tensile Test Device.* American Association of State Highway and Transportation Officials.
2. Greaves et al, (2011). *Poisson's ratio and modern materials*. Nature Materials 10, 823-837.b4e
3. Hervé Di Benedetto, Brice Delaporte and Cédric Sauzéat (2007). "Three-Dimensional Linear Behavior of Bituminous Materials: Experiments and Modeling." *Int. J. Geomech.* 7, SPECIAL ISSUE: Constitutive Modeling and Computational Analysis of Asphalt Pavements, 149-157.

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